

Instructions for dear students,

1. Please submit your assignment before the designated submission date. Late submissions will not be considered for grading.
2. Create separate answer sheets for this assignment. Remember to collect your submitted sheets once the assignment has been checked.
3. This assignment contributes a total of 20 marks to the internal marks evaluation criteria outlined by Pokhara University.
4. If you encounter any issues, feel free to consult with your instructor. You can either email your queries to the official email provided or visit the Department of CE/IT/SW Engineering at Everest Engineering College in person.

Last date of submission: 14th February 2025

Chapter – 01: Basic Discrete Structures

1. Define set, subset, power set, and universal set with examples. Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using set identities.
2. Explain the Inclusion-Exclusion Principle with an example. The bit strings for a universal set $\{0,1,2,3,4,5,6,7,8,9\}$ are $A = 110\ 010\ 1110$ and $B = 101\ 101\ 1100$, respectively. Use bit strings to find the $A \cup B$, $A \cap B$, $A - B$ and $\bar{A}$ of the sets.
3. Define and differentiate between Injective, Surjective, and Bijective functions with examples. Prove whether the function $f(x) = x^2 + 2x + 1$ is injective, surjective or bijective for $x \in \mathbb{R}$.
4. Explain the Ceiling Function and Floor Function with real-life applications in computer science. If $f(x) = 2x + 3$ and $g(x) = x^2$, find the composite function $(f \circ g)(x)$ and $(g \circ f)(x)$.
5. What is a Boolean function? State and prove De Morgan's law using truth table.
6. Prove that if a number is divisible by 6, then it is also divisible by 2 and 3.
7. Find the Greatest Common Divisor (GCD) of 252 and 198 using the Euclidean Algorithm.
8. Define unit matrix, diagonal matrix and unit matrix. Given the matrices:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Find AND, OR and Product of the matrices.

9. Explain the Pigeonhole Principle and prove that among 7 people, at least two must have been born on the same day of the week.
10. A password consists of 3 letters followed by 2 digits. How many such passwords can be formed if:
 - a. Repetition of letters and digits is allowed?
 - b. Repetition is not allowed?

Chapter – 02: Relations

1. Let R be a relation on a set $A = \{a, b, c\}$ defined as: $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$ Check whether R is Reflexive, Symmetric, and Transitive.
2. Represent the following relation as a relation matrix and directed graph (digraph):
$$R = \{(1, 2), (2, 3), (3, 4), (4, 1), (2, 2)\} \text{ on } A = \{1, 2, 3, 4\}$$
3. What is the difference between transitive closure and reflexive closure? Explain with an example. Construct transitive closure for following relation with Warshall's Algorithm:

$$R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4. Differentiate between equivalence relations and partial order relations with suitable examples. Consider the set $A = \{2, 4, 8, 16\}$ with the relation divides (i.e., xRy if and only if x divides y).
 - a. Draw its Hasse diagram.
 - b. Show whether the relation is reflexive, antisymmetric, and transitive.

Chapter – 03: Recurrence Relation

1. Find the complete solution of the recurrence relations:
 - a. $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 2$, $a_1 = 3$
 - b. $a_n = 4a_{n-1} - 3a_{n-2}$, $a_0 = 1$, $a_1 = 2$
 - c. $a_n = 4a_{n-1} - 6a_{n-2} + 2n$, $a_0 = 1$, $a_1 = 2$
 - d. $a_n = 4a_{n-1} - 3a_{n-2} + 5^n$, $a_0 = 3$, $a_1 = 1$
2. Find the explicit formula for the Fibonacci sequence.
3. Solve TOH problem using iterative approach.

Chapter – 04: Logic Induction and Reasoning

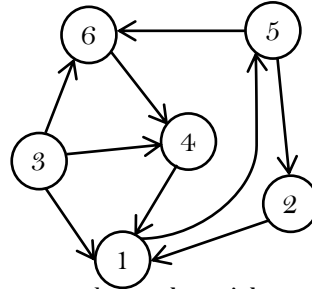
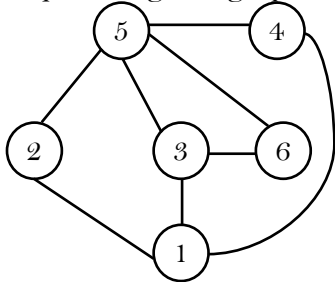
1. Define proposition with example. Construct the truth table for the following expressions:
 - a. $(P \vee Q) \wedge \neg P$
 - b. $P \Rightarrow Q$
2. What is propositional logic? How is it different from predicate logic? Write the symbolic form of the following statement:
 - a. "If it is raining, then the ground is wet."
 - b. "If the computer is on, then the monitor will display an image."
 - c. "Either I will watch a movie or read a book, but not both."
 - d. "You will pass the exam if and only if you study hard."
 - e. "If a student studies hard and attends classes, then they will pass the exam or get a scholarship."
 - f. "If the system is not secure or the user is careless, then data will be lost."
 - g. "A person is eligible to vote if and only if they are a citizen and at least 18 years old."
 - h. "If a user enters the correct password and has security clearance, they can access the system; otherwise, access is denied."
3. What is a tautology and a contradiction? Construct the truth table and determine if the following proposition is a tautology:

$$(P \vee Q) \wedge (\neg P \Rightarrow Q)$$
4. Define predicate logic, universal and existential quantifiers?
 - a. "All students in the class submitted their assignments on time."
 - b. "Every teacher has at least one student who admires them."
 - c. "For every person, there exists a book that they like."
 - d. "There is a doctor who treats every patient."
 - e. "Not every student passed the exam."
 - f. "If a person is a programmer, then they know at least one programming language."
 - g. "There is exactly one president of the country."
 - h. "If every person loves chocolate, then some people love ice cream"

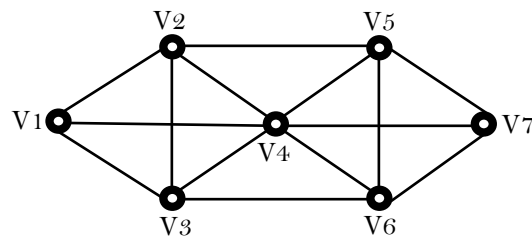
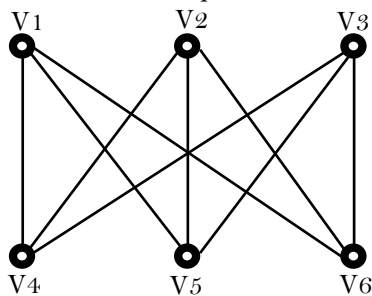
5. Explain the rules of inference: Modus Ponens, Modus Tollens, Hypothetical Syllogism, and Disjunctive Syllogism with examples. Using rules of inference show that given premises lead to conclusion statement.
- a. Premises:
- a. "If it snows, then roads become slippery."
 - b. "If roads become slippery, then driving is dangerous."
 - c. "It is snowing today."
- Conclusion:
"Driving is dangerous today."
- b. Premises:
- a. "If a book is available in the library, then students can borrow it."
 - b. "The book 'Discrete Mathematics' is available in the library."
 - c. "Alice is a student."
 - d. "If a student can borrow a book, they can read it at home."
- Conclusion:
"Alice can read 'Discrete Mathematics' at home."
- c. Premises:
- a. "If a device is not charged, it will turn off."
 - b. "The laptop is not charged."
 - c. "If the laptop turns off, the work will not be saved."
- Conclusion:
"The work will not be saved."
- d. Premises:
- a. "If a computer is slow, it needs more memory."
 - b. "If a computer needs more memory, the user must buy a new RAM chip."
 - c. "The computer is slow."
 - d. "The user has not bought a new RAM chip."
 - e. "If the user has not bought a new RAM chip, the computer will continue to be slow."
- Conclusion:
"The computer will continue to be slow."
6. What is the difference between a direct proof and an indirect proof? Prove using direct proof:
- a. Sum of two odd number is even.
 - b. Square of odd number is also odd.
7. Differentiate between proof by contraposition and proof by contradiction. Prove the given statements:
- a. If n^2 is even then n is even using contraposition.
 - b. $\sqrt{2}$ is irrational using contradiction.
8. State the principle of mathematical induction. Use mathematical induction to prove:
- a. Sum of n number is $\frac{n(n+1)}{2}$
 - b. $2^n > n^2$ for $n \geq 5$
9. Explain the difference between weak induction and strong induction with an example. Use strong induction to prove that every integer $n \geq 2$ can be written as a product of prime numbers.

Chapter – 05: Graph Theory

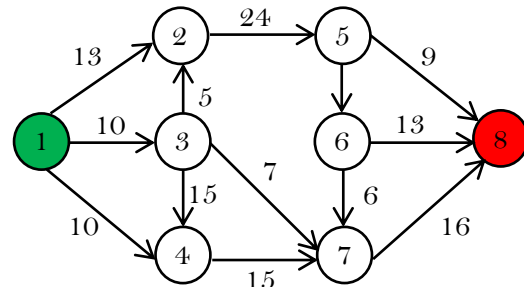
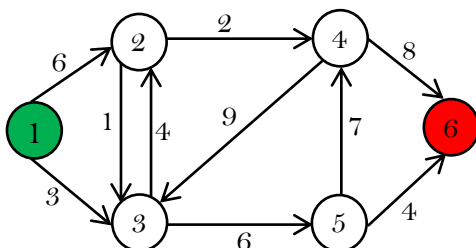
- Define the terms with example:
 - Directed and undirected graph
 - Simple, multi and pseudograph
 - Weighted graph
 - Cyclic and acyclic graph
 - Walk, trail, path and circuit
- Represent given graph into Adjacency matrix, incidence matrix and path matrix:



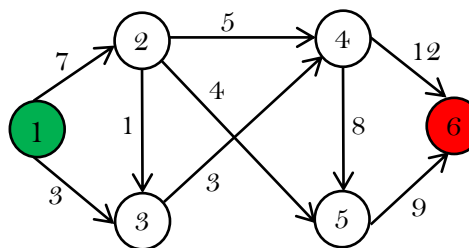
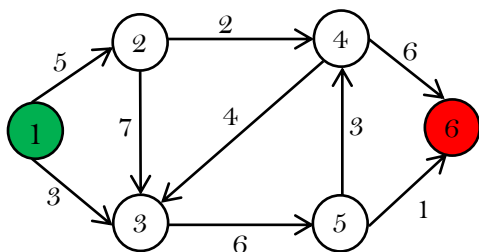
- Define regular, complete, cycle, and connected graphs with examples.
- What is a bipartite graph? Prove that a complete bipartite graph does not have a Eulerian circuit.
- For following graphs determine whether or not there exists: a Euler path, a Euler cycle, a Hamiltonian path or a Hamiltonian cycle.



- Determine if the graph is Hamiltonian graph for given:
 - A graph with 10 vertices, where each vertex has a degree of at least 5,
 - A graph G with 7 vertices, where for every pair of non-adjacent vertices u and v , the sum of their degrees is at least 7.
- Given directed graphs with edge capacities, calculate the maximum flow using the Ford-Fulkerson algorithm:



- Explain the Max-Flow Min-Cut theorem with an example. Determine the max flow of given graph using Max-Flow Min-Cut theorem:



Chapter – 06: Language Grammar and Automata

1. Define language and grammar. Explain Chomsky's hierarchy of grammars.
2. Define Finite automata and DFA. Construct DFA that recognizes the language:
 - a. $L = \{w \mid w \text{ starts with '10'}\}$.
 - b. $L = \{w \mid w \text{ contains the substring '110'}\}$.
 - c. $L = \{w \mid w \text{ ends with '11'}\}$.
 - d. $L = \{w \mid w \text{ has an even number of 1s}\}$.
 - e. $L = \{w \mid w \text{ is divisible by 3 when interpreted as binary numbers.}\}$
3. Differentiate between deterministic and non-deterministic finite automata. Construct NFA that recognizes the language:
 - a. $L = \{w \mid w \text{ starts with '0' or ends with '1'}\}$.
 - b. $L = \{w \mid w \text{ contains exactly one occurrence of '01'}\}$.
 - c. $L = \{w \mid \text{the second symbol from the right is '1'}\}$.
4. Convert the NFA constructed in QN. 3 into DFA.
5. Define regular expressions. Construct a regular expression for automata constructed at QN. 2(b), 2(c), 2(e) and 3(c).
6. Convert the following regular expression into a finite automaton:
 - a. $R = (0+1)101(0+1)^*$
 - b. $R = 0^*1(1+0)0^+$